# Rabiul Awal

Department of Computer Science and Operations Research
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Mila - Quebec AI Institute, QC, Canada

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**Research Interests:** Generative Models, World Models, Reinforcement Learning

## Current Position

*09/2024 – present* **Mila - Quebec AI Institute, Montreal**
Graduate Student Researcher

## Education

*09/2024 – present* **University of Montreal**
Ph.D. in Computer Science
Department of Computer Science and Operations Research
Advisor: Aishwarya Agrawal
Courses: Reinforcement Learning and Optimal Control, Vision and Language
CGPA: 4.15/4.3

*01/2021 – 12/2022* **University of Saskatchewan**
MSc. Computer Science
Advisor: Michael Horsch, Roy Ka-Wei Lee
CGPA: 88/100

*03/2014 – 12/2018* **Noakhali Science and Technology University, Bangladesh**
BSc. Computer Science

## Previous Experience

*05/2026 – 09/2026* **Qualcomm Research, Amsterdam**
Research Intern

*07/2024 – 09/2025* **ServiceNow Research, Montreal**
Visiting Researcher in the Multimodal Team
Advisor: Sai Rajeswar

*01/2023 – 08/2024* **Mila - Quebec AI Institute, Montreal**
Research Intern
Advisor: Aishwarya Agrawal

*06/2020 – 11/2022* **Singapore University of Technology and Design**
Student Researcher at Social AI Studio
Advisor: Roy Ka-Wei Lee

*12/2019 – 12/2020* **Goava Sales Intelligence AB, Dhaka**
Software Engineer

## In-Progress Research

[1] *Feature-Space Consistency Improves Flow Maps Training* Introduces a feature-space consistency objective to accelerate training and improve the stability of one-/few-step diffusion models.
https://drive.google.com/file/d/1tt0go1c69HqtN3LTCYLwY_RtD0HzMhqW/

[3] *Denoising Embedding Models* Developing a unified denoising framework that serves as both a self-supervised visual encoder and an image generator, while learning a generative embedding space.
https://drive.google.com/file/d/1aoZ66d-IHze9Kn6lJ1Y5XsoEib6jcucG/

## Peer-Reviewed Workshop Publications (Non-Archival)

[W2] **Rabiul Awal**, Jinseong Jeong, Ankur Sikarwar, Parisa Kordjamshidi, Andrii Zadaianchuk, Sai Rajeswar, Paul Hongsuck Seo, Aishwarya Agrawal. One Flow Transformer for Imagination and Control. In *Model-Based RL in the Era of Generative World Models Workshop at the Reinforcement Learning Conference* (*RLC*). 2026.

[W1] **Rabiul Awal**. Dynamics Forcing for Diffusion World Models. In *Model-Based RL in the Era of Generative World Models Workshop at the Reinforcement Learning Conference* (*RLC*). 2026.

## Peer-Reviewed Conference Publications

Google Scholar for full list:
https://scholar.google.com/citations?user=oOcLfUMAAAAJ&hl=en
*Denotes equal contribution

[C8] Aarash Feizi, Shravan Nayak, Xiangru Jian, Kevin Qinghong Lin, Kaixin Li, **Rabiul Awal** + 11 others. Grounding Computer Use Agents on Human Demonstrations. In *International Conference on Learning Representations* (*ICLR*). 2026.

[C7] Juan A. Rodriguez, Haotian Zhang, Abhay Puri, Aarash Feizi, Rishav Pramanik, Pascal Wichmann, Arnab Mondal, Mohammad Reza Samsami, **Rabiul Awal** + 7 others. Rendering-Aware Reinforcement Learning for Vector Graphics Generation. *Conference on Neural Information Processing Systems* (*NeurIPS*). 2025.

[C6] Saba Ahmadi*, **Rabiul Awal***, Ankur Sikarwar*, Amirhossein Kazemnejad*, Ge Ya Luo, Juan A Rodriguez, Sai Rajeswar, Siva Reddy, Christopher Pal, Benno Krojer, Aishwarya Agrawal. The Promise of RL for Autoregressive Image Editing. *Conference on Neural Information Processing Systems* (*NeurIPS*). 2025.

[C5] Anikter Didolkar*, Andrii Zadaianchuk*, **Rabiul Awal***, Maximilian Seitzer, Efstratios Gavves, Aishwarya Agrawal. CTRL-O: Language-Controllable Object-Centric Visual Representation Learning. *IEEE / CVF Computer Vision and Pattern Recognition Conference* (*CVPR*). 2025.
**Spotlight presentation** at Multimodal Algorithmic Reasoning Workshop, CVPR'25.

[C4] **Rabiul Awal**, Mahsa Massoud, Aarash Feizi, Zichao Li + 9 others. WebMMU: A Benchmark for Multimodal Multilingual Website Understanding and Code Generation. *The Conference on Empirical Methods in Natural Language Processing* (*EMNLP*). 2025.
**Spotlight presentation** at Multimodal Algorithmic Reasoning Workshop, CVPR.

[C3] **Rabiul Awal***, Saba Ahmadi*, Le Zhang*, and Aishwarya Agrawal. VisMin: Visual Minimal-Change Understanding. *The Conference on Neural Information Processing Systems* (*NeurIPS*). 2024.
**Contributed talk** at SpLU-RoboNLP Workshop, ACL 2024.

[C2] Shravan Nayak, Kanishk Jain, **Rabiul Awal**, Siva Reddy, Sjoerd van Steenkiste, Lisa Anne Hendricks, Karolina Stanczak, Aishwarya Agrawal. Benchmarking Vision Language Models for Cultural Understanding. *The Conference on Empirical Methods in Natural Language Processing* (*EMNLP*). 2024. **Oral Presentation**.

[C1] Le Zhang, **Rabiul Awal**, Aishwarya Agrawal. Contrasting Intra-modal and Ranking Cross-modal Hard Negatives to Enhance Visio-linguistic Compositional Understanding. *IEEE / CVF Computer Vision and Pattern Recognition Conference* (*CVPR*). 2024. **Spotlight presentation** at the Open-Domain Reasoning Multi-Modal Workshop, CVPR'23.

## Selected Awards, Grant & Fellowship

*2026* Fonds de recherche du Québec – Nature et technologies (FRQNT) Scholarship (25,000$).

*2026* DIRO Excellence Scholarship for the Winter 2026 session (3900$)

*2025* DIRO Excellence Scholarship for the Fall 2025 session 2025-2026 (3000$)

*2025* Outstanding Reviewer at CVPR

*2024* Selected for CIFAR DLRL Summer School at Toronto

*2024* Selected for Upper Bound Talent Bursary at Alberta ($1500)

*2022* Selected for Alberta University AI Week Talent Bursary ($1500)

## Service

### Broader Community

*2024–present* **Supervision Request Process Evaluation Committee**
Help professors recruit (pre-evaluation of application) msc and phd students at Mila

*2025–present* **Whiteboard Session Lead Organizer**
Help organize the RL for Foundation Models reading group for internal lab at Mila

*2021–2022* **Computer Science Graduate Council MSc Representative**
Helped in strengthening graduate research activities, hosting social events for grad students at University of Saskatchewan

*2023–2024* **Language Grounding Reading Group Organizer**
Help supporting running the reading group on vision language at Mila

### Research Community

*02/2026* **Workshop Organizer at Mila**
World Modeling Workshop

*05/2025* **Workshop Organizer at CVPR25**
Vision Language Models For All: Building Geo-Diverse and Culturally Aware Vision-Language Models

*2021–present* **Reviewer**
AAAI26, NeurIPS25, CVPR25, ECCV24, CVPR24, ACL23, SIGIR22, AACL22, ICDM21, SIGKDD21